



POLITECNICO
MILANO 1863



Network Data Analysis Laboratory

Proposed projects

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General requirements (1/2) – Projects tasks

- All projects **must** include the following main steps:
 1. Raw data visualization/analysis
 - Does data suggest anything (ML algorithm, promising features, redundancy, ...)?
 2. Data preprocessing (e.g., generate «ordered» samples from raw data)
 3. ML models optimization and training
 - Hyperparameters tuning with cross-validation
 4. Testing the performance and evaluating different scenarios (**see specific projects assignment**)



General requirements (2/2) – Methodology

- All projects **must** include (for 12 points)
 - Specific project assignment (see later)
 - 2 or 3 ML algorithms (usually recommended in the assignment)
 - Different performance metrics: MSE, MAE, Accuracy, Precision, Recall, F-score, training duration, ...
- Additionally, projects **may** include (at least) one "advanced" task (for 3 points)
 - Transfer learning: train on one domain, test on another domain (e.g., different datasets, different tasks, etc.)
 - Federated learning: compare "global" models (all data available at one location) vs. "local" models that share knowledge through Federated Learning
 - Explainability: apply XAI (eXplainable Artificial Intelligence) frameworks to interpret/explain model reasoning and validate the model
 - Project-specific advanced task



Projects list

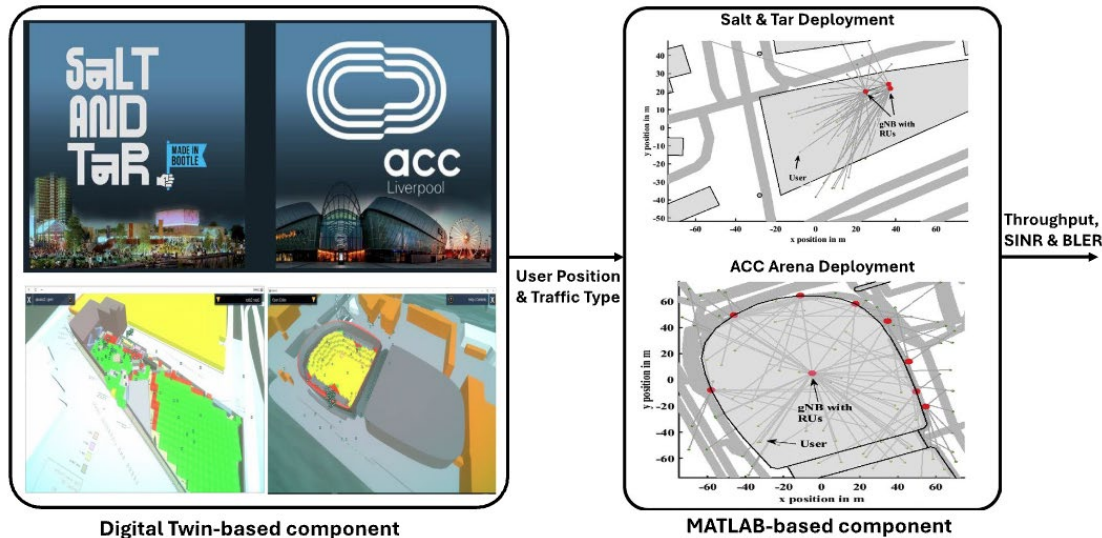
1. Packet Loss Event Classification with TL
2. Packet Loss Event Classification with FL
3. Weather-based performance prediction of satellite-internet provider
4. What Network Am I Using? RAT Classification with TL
5. What Network Am I Using? RAT Classification with FL
6. YouTube Streaming: Will I Experience QoE Variation?
7. Throughput Prediction in a Dense 5G deployment with FL
8. Throughput Prediction in a Dense 5G deployment with TL
9. Throughput Prediction in a Dense 5G deployment with Vertical FL
10. IXP Traffic Forecasting (anomaly detection) with TL
11. IXP Traffic Forecasting (traffic volume) with TL
12. IXP Traffic Forecasting (time-series forecasting) with FL
13. QoT forecasting with TL
14. EDFA profile MIMO regression
15. EDFA profile MISO regression
16. EDFA profile regression comparing XGBoost and KAN
17. Synthetic data generation in microwave networks (SMOTE vs VAE)
18. Synthetic data generation in microwave networks (SMOTE vs GAN)
19. Optical failure localization: single-step multi-class classification
20. Optical failure localization: two-step multi-class classification



Project #7-8-9 – Throughput Prediction in a Dense 5G deployment

Dataset [1]

- **Source [2]:** Simulated 5G dense deployment scenarios in two different Liverpool venues. The venues in consideration are Salt&Tar (3k users) and ACC Arena (12k users). The dataset has been experimentally validated.
- **Data Scope:** 10k samples per user per venue (Salt&Tar sampled every second while ACC Arena sampled every three seconds) collected both user terminal-specific and radio-unit-specific.
- For each sample, data includes timestamp, user ID, user position (3D coordinates), traffic type¹, RU association², downlink SINR (dB), uplink SINR (dB), throughput (Mbps)³, PRB⁴, and BLER⁵.



¹ Defined in six classes (0: 'off', 1: 'idle', 2: 'constant rate', 3: 'video', 4: 'gaming', 5: 'http')

² Specific 5G Radio Unit (RU) that the user is connected to (highest received signal strength)

³ Not specified if uplink, downlink or combined.

⁴ PRB (Physical Resource Blocks): Utilization of physical resource blocks scheduled for the user's transmission or reception.

⁵ BLER (Block Error Rate): Rate of blocks received with errors for that user.

[1] Maheshwari, Mukesh Kumar, et al. "5G High Density Demand (HDD) Dataset in Liverpool City Region, UK (Supplement)." (2025).

[2] Maheshwari, Mukesh Kumar, et al. "5G High Density Demand Dataset in Liverpool City Region, UK." Scientific data (2025).



Project #7-8-9 – Throughput Prediction in a Dense 5G deployment

Assignment

- Assignment: Can we predict the user throughput given user terminal and radio unit features?
 - (12 points) Perform throughput regression on ACC Arena users
 - Compare two models (Neural Network and Random Forest)
 - Use the following features:
 - BLER, PRB, SINR, user traffic values and ...
 - Team 7: ...the throughput of operator measurement devices ($ID=[0,X-1]$) that have the same traffic type as the current user (Note: These X users must be excluded from the dataset used for prediction)
 - Team 8: ...the features from X closest users
 - Team 9: ...the number of users within X meters from the target device
- } Experiment different values of X
- (3 points) Advanced task
 - Team 7: Simulate a **Federated Learning** (FL) approach over users with FedAvg. Compare with single device train set performance
 - Team 8: **Transfer Learning** from ACC Arena to Salt&Tar scenario. Compare with the same model trained on a limited Salt&Tar train set
 - Team 9: **Vertical FL** splitting features between the user terminal and the radio unit using a combined Linear Regressor for throughput prediction. Compare the results with more complex models (e.g., a Neural Network) trained only with single device features

